MD AMIR KHAN

Applied AI Engineer · LLM Systems & Quantitative Analytics

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Summary

Build production **LLM systems** — structured extraction, agentic RAG, multi-model inference — and the **ML forecasting stack** that consumes them, running daily on the DXT Commodities gas & power trading desk.

Education

Stevens Institute of Technology — *Hoboken, NJ, USA*
Master of Science in Financial Engineering & Analytics (STEM-designated)

North South University — *Dhaka, Bangladesh*
Bachelor of Business Administration · Major: Finance · Minor: Mathematics

Experience

DXT Commodities North America — *Stamford, CT*
Applied AI Engineer & Quantitative Analyst — Market Fundamentals (LNG & Power) — *March 2026 – Present*

- **Pipeline transparency platform — LLM structured extraction at production scale.** Built the scraping and LLM extraction pipeline reading maintenance and capacity notices from 19 U.S. interstate gas pipelines across 8 different portal architectures (15+ format-specific scrapers: HTML, PDF, Excel, ASP.NET with TLS impersonation for WAF-protected sites). **Structured extraction on the Claude API** turns free-form notices into typed capacity-impact records with a caching layer that cuts LLM calls on unchanged input. FastAPI + SQL Server; tracks 40+ active and 198+ upcoming restrictions on a peak day.
- **WattsApp — full-stack PJM analytics with natural-language query.** Full-stack platform covering **4,305 units / 232.5 GW nameplate** across 2,262 plants in 13 states + DC. Overview, thermal, non-thermal, plants, insights, and an **LLM-powered natural-language query view** used daily by the power desk. Next.js + React + TypeScript, FastAPI, SQL Server, Docker.
- **U.S. LNG feed-gas forecasting.** End-to-end forecasting pipeline covering **126 export trains at 13 terminals (~22,600 MMcf/d nameplate)**. Three-model validation framework; **under 3% MAPE** on an 8-month out-of-sample holdout.
- **U.S. natural-gas demand forecasts.** 48 models — 12 regions × 4 end-use sectors — producing daily 30-day forecasts for the Lower 48. XGBoost for weather-sensitive sectors, Ridge for the slower ones; walk-forward cross-validation. Driven by NWS temperature forecasts and Fed industrial-production data, trained on EIA state-level consumption.
- **Wattson — multi-ISO power-price analytics.** Live platform for PJM, ERCOT, NYISO, MISO, and ISO-NE: historical LMPs by energy / congestion / marginal-loss component, forward-contract settlements with strip aggregation, and a Price ⇄ Heat-Rate toggle connecting power and gas markets.
- **Real-time gas-production model & Permian Basin intelligence.** OLS/Ridge scaling framework combining licensed pipeline-nominations data with EIA-914 to produce a current-month U.S. production estimate inside the two-month EIA reporting lag. Permian codebase covers daily production, egress across the four major pipelines, and Waha basis pricing.
- **Pipeline Force Majeure alerts & PEPCO nodal-basis screen.** Real-time alerting service polls electronic bulletin boards across **32 U.S. interstate gas pipelines** every 5 minutes and pushes notices to Microsoft Teams (response time: hours → minutes). PEPCO FTR-bidding screen covers **~8.77 M hourly observations** across 61 nodes and 8+ years of PJM day-ahead settlement data.

Stevens Institute of Technology — *Hoboken, NJ*
Research Assistant — LLM-Generated Views in a Black–Litterman Portfolio (MSc Thesis) — *January 2024 – December 2025*

- Designed and built an **LLM-powered forecasting system** treating language models as queryable, uncertainty-aware forecasters rather than static tools — a case study in structured output extraction, multi-model evaluation, and confidence-aware system design.
- Built a **multi-LLM inference pipeline** (GPT-4o-mini, LLaMA-3.1, Gemma, Qwen-2) using **vLLM** for high-throughput serving; structured prompts convert price, sector, and news data into scalar return forecasts.
- Solved the **uncertainty-quantification problem** by repeated querying and treating output dispersion as a Bayesian confidence signal — sample mean fed as the view, sample variance as view uncertainty, into a downstream **Black–Litterman** update rather than being discarded.
- Full backtesting pipeline: 50 largest S&P 500 names, **Feb 2024 – Jan 2025**, biweekly rebalancing, benchmarked against 3 baseline strategies; **best model (LLaMA-3.1) beat the S&P 500 benchmark — Sharpe 1.29 vs. 0.87**.

Stevens Institute of Technology — *Hoboken, NJ*
Quantitative Research Assistant — School of Business — *April 2025 – February 2026*

- Implemented the **ε-subdivision Robust PCA framework for dynamic factor portfolios** in Python with Prof. Papa Momar Ndiaye — block decomposition of the eigenspectrum against a tolerance ε, Gram–Schmidt construction of the closest orthonormal

basis to the prior period's factors, and rupture-detection that resets factor tracking when block-mean eigenvalues shift beyond a threshold δ. Added K-Means clustering on eigenvector features as an independent signal of factor-structure change.
- Validated on ∼6.5 years of daily returns across the 11 S&P 500 GICS sector ETFs, spanning the COVID-19 shock and the post-pandemic inflation cycle. Fed robust and standard covariance estimates into a mean-variance optimizer; the robust approach stabilized factors at portfolio volatility essentially identical to standard PCA.

Teaching Experience

Stevens Institute of Technology *Hoboken, NJ*
Teaching Assistant — Quantitative Portfolio Theory (Financial Engineering program) *September 2025 – December 2025*

Technical Skills

AI / LLM Engineering:	LangChain · LangGraph · LangSmith · RAG (HyDE, CRAG, Self-RAG, Graph RAG, Agentic RAG) · LLM Agents & Subagents · Structured Output Extraction · Prompt Engineering · Multi-LLM Serving (vLLM) · Anthropic Claude & OpenAI APIs · Model Context Protocol (MCP)
Machine Learning:	Scikit-learn · XGBoost · Ridge / OLS · Statsmodels · Pandas · NumPy · Time-Series Forecasting · Walk-Forward Cross-Validation · Robust PCA · Black–Litterman
Full-Stack & Infra:	FastAPI · Next.js · React · SQLAlchemy · SQL Server · Docker · AWS · REST APIs · Git / GitHub / GitLab
Languages:	Python · SQL · TypeScript · JavaScript
Domain (Energy):	U.S. Natural Gas & Power Fundamentals · LNG Feed-Gas · Pipeline Capacity Analysis · ISOs / RTOs (PJM, ERCOT, NYISO, MISO, ISO-NE) · Wholesale LMPs · FTR Markets · Heat-Rate Analysis

Certifications & Awards

- **Anthropic Education — Certified Track** (Anthropic, 2026): Claude API, Claude Code, Model Context Protocol (Intro + Advanced), Agent Skills, Subagents.
- **Advanced RAG** (CampusX, 2026) · **AI Engineer Bootcamp 2026: LLMs, RAG, Agents, Vector DBs** (Udemy, 2026) · **Vector Databases: Fundamentals to Production** (Udemy, 2026).
- **Full-Stack Web Track** (Modern JavaScript, TypeScript, React, Next.js, Node/Microservices, Full-Stack Bootcamp; Udemy, 2026) · **Developer Foundations** (Linux CLI, Git & GitHub; Udemy, 2026).
- **Probability — The Science of Uncertainty and Data** (MIT / edX, 2022) · **Python and Statistics for Financial Analysis** (Coursera, 2022) · **Akuna Capital Options 101** (2025) · **Algorithmic Trading with Python, ChatGPT, ML** (Udemy, 2025) · **FastAPI Complete Course** (Udemy, 2025).
- Student Member, **CFA Society New York**.